

MATB24H3

Exercise Questions

Student Name: Your Name

Student Number: Your Student Number

2.1.1. Consider the set $X = \mathbb{Q} \setminus \{0\}$ and the map $\star$ is depicted by $a \star b = \frac{a}{b}$. Is $\star$ an associative binary operation?

Solution:

2.1.2. Is $(\mathcal{P}(\mathbb{N}), \cap)$ a monoid? what is the identity element? How about $(\mathcal{P}(X), \cup)$ for any arbitrary set X ?

Solution:

2.2.1. Suppose d is square-free, so there is no integer m such that $m^2 = d$. Show that $(\mathbb{Q}(\sqrt{d}), +, \cdot)$ forms a field. What happens if $d = m^2$ for some m ?

Solution:

2.2.2. Now, can we extend a finite field in the same way we extended an infinite field? If so, give an example.

Solution:

2.2.3. Prove that $0_{\mathbb{F}}$, $1_{\mathbb{F}}$ and the additive inverse element $-a$ of any element $a \in F$ are unique.

Solution:

2.2.4. Prove $(-a)(-b) = ab$ for all $a, b \in \mathbb{F}$.

Solution:

2.2.5. Prove that the triple $(\mathbb{Z}_p, + \bmod p, \cdot \bmod p)$ is a field $\iff p$ is a prime. Use the fact that if two integers m, n are relatively prime, then there exist integers r and s such that $mr + ns = 1$.

Solution:

2.2.6. Prove that the 2-tuple $(S, \circ)$, where S is the set of rotational symmetries of a square and $\circ$ is the composition operation, does indeed form a group. Moreover, find the symmetry group of a rectangle.

Solution:

2.2.7. Characterise all idempotent elements of $(\mathcal{M}_{2 \times 2}(\mathbb{R}), +, \cdot)$.

Solution:

2.2.8. Prove that $\mathcal{C}(\mathcal{R})$ is a subring of any ring $(\mathcal{R}, +, \cdot)$.

Solution:

2.2.9. Let $(\mathcal{R}, +, \cdot)$ be a ring and suppose that $\alpha \in \mathcal{R}$ has a multiplicative inverse. Moreover, suppose that $\alpha \in \mathcal{C}(\mathcal{R})$. Is $\alpha^{-1} \in \mathcal{C}(\mathcal{R})$?

Solution:

2.2.10. Does the set of functions with infinitely many roots form a subring of $\mathcal{F}(\mathbb{R})$? Justify your answer.

Solution:

2.2.11. There is no function $g \in \text{supp}(\mathcal{F}(\mathbb{R}))$ that acts as multiplicative identity for all elements in $\text{supp}(\mathcal{F}(\mathbb{R}))$. Can you prove this?

Solution:

2.2.12. Determine the zero and the identity elements of the polynomial ring $(\mathcal{R}[x], +, \cdot)$ where $\mathcal{R} = \mathcal{C}(\mathcal{M}_{2 \times 2}(\mathbb{R}))$.

Solution:

2.2.13. True/False. Justify:

1. Let $\mathbb{F}$ be any field. Suppose that α and β are two elements of an extension field $\mathbb{K}$ of $\mathbb{F}$. Then $\mathbb{F}(\alpha)$ and $\mathbb{F}(\beta)$ are always distinct extension fields of $\mathbb{F}$.
2. Given any set X , there is no operation $\cdot$ that can be defined on $[X]^{\leq |X|}$ that makes it a group.

3. Given a monoid $(\mathcal{S}, *)$ there exists a binary operation $\cdot$ on $\mathcal{S}$ such that $(\mathcal{S}, \cdot)$ no longer admits an identity element. If so, what assumption did you need to make?
4. $(\mathbb{R}, \cdot)$ forms a group.
5. $(\mathbb{R}, +)$ forms a group with the identity element being 1.
6. Suppose $(\mathbb{F}, +, \cdot)$ is the field. Then $(\mathbb{F}, +)$ and $(\mathbb{F}, \cdot)$ both form a group.

Solution:

3.1.1. Had we defined vector addition by $\hat{A} \boxplus \hat{B} = [a_{ij} \cdot b_{ij}]$, i.e., pointwise multiplication of the entries in Example 3.1.3, would $(\mathcal{M}_{m \times n}, \boxplus, \odot, (\mathbb{F}, +, \cdot))$ form a vector space over $\mathbb{F}$?

Solution:

3.1.2. Prove properties 2, 4 and 5 of Theorem 3.1.2, namely, for any vector space $V = (V, \boxplus, \odot, (\mathbb{F}, +, \cdot))$ over field $\mathbb{F}$:

2. The additive inverse of any $v \in V$ is unique.
4. Any scalar of the zero vector yields the zero vector, i.e. $\forall \alpha \in \mathbb{F}, \alpha \odot \vec{0} = \vec{0}$.
5. For all $v \in V$ and $c \in \mathbb{F}$, $(-c) \odot v = c \odot (-v) = -1 \odot (c \odot v)$.

Solution:

3.2.1. Prove Theorem 3.2.2, namely if $S \subseteq V$ then $\text{span}(S) \subseteq V$ is a subspace of V .

Solution:

3.2.2. Prove Theorem 3.2.3, namely: Let J be any index set (It is a set whose elements are used to enumerate the elements of another set.) and let $\{W_\alpha \mid \alpha \in J\}$ be a collection of subspaces of V . Then $\bigcap_{\alpha \in J} W_\alpha$ is a subspace of V .

Solution:

3.2.3. Prove Theorem 3.2.4, namely: Let $W_1 \subseteq W_2 \subseteq W_3 \subseteq \dots$ be an increasing chain of subspaces of V . Then $\bigcup_{n \in \mathbb{N}} W_n$ is a subspace.

Solution:

3.3.1. Show that the set $\{e^x, e^{2x}, e^{3x}\}$ is linearly independent in the vector space of $\mathcal{F}(\mathbb{R})$ over $\mathbb{R}$.

Solution:

3.4.1. Prove Proposition 3.4.1, namely, the following:
Let $T: V \rightarrow W$ be a linear transformation from V to W . Then:

1. For all $x, y \in V$, $T(x - y) = T(x) - T(y)$
2. T maps the zero vector $\vec{0}_V$ of V to the zero vector $\vec{0}_W$ of W .
3. If $T': W \rightarrow Z$ is also a linear transformation, their composition $T' \circ T$ is also a linear transformation from V to Z where $T' \circ T = T'(T(v))$. So for every vector $\vec{v} \in V$, we have $(T' \circ T)(\vec{v}) := T'(T(\vec{v}))$.
4. If V_1 is a subspace of V and W_1 is a subspace of W , then $T[V_1]$ is a subspace of W and $T^{-1}[W_1]$ is a subspace of V .

Solution:

3.4.2. Prove

$$T: V \rightarrow W \text{ is injective} \iff \ker T = \{\vec{0}\}$$

Solution:

3.4.3 $(\mathcal{L}(V, W), \boxplus, \odot, (\mathbb{F}, +, \cdot))$ forms a vector space over $\mathbb{F}$, where $\boxplus$ and $\odot$ are the pointwise addition of linear maps and scalar multiplication with the linear map in W , respectively.

That is given $T, S \in \mathcal{L}(V, W)$ and $\alpha \in \mathbb{F}$

$$T \boxplus S := T \underbrace{+}_W S$$

$$\alpha \odot T := \alpha \underbrace{\cdot}_W T$$

So, given any $v \in V$ the image of v under the linear maps $T \boxplus S$ and $\alpha \odot T$ would be $T(v) +_W S(v)$ and $\alpha \cdot_W T(v)$ respectively.

That is, $\mathcal{L}(V, W)$ is closed under both $\boxplus$ and $\odot$. That is to show that $T \boxplus S$ and $\alpha \odot T$ are indeed linear transformations from V to W and all the axioms of a vector space are satisfied.

Solution:

3.4.4. Prove:

Let V and W be two vector spaces over $\mathbb{F}$ with $\dim(V) = n$ and $\dim(W) = m$. Then, $\dim(\mathcal{L}(V, W)) = mn$

Hint: Note that a linear transformation T is determined by its action on its basis elements of its domain. So, if $\mathcal{B}$ is a basis for V over $\mathbb{F}$, and $T \in \mathcal{L}(V, W)$, then as long as we know what $T(b)$ is for every $b \in \mathcal{B}$, (the basis could be finite or infinite) we could then determine the image of an arbitrary vector $v \in V$. Since $v = \sum_{i=1}^n \alpha_i b_i$ for some $n \in \omega$, $b_i \in \mathcal{B}$, and $\alpha_i \in \mathbb{F}$, then by the properties of linear operators we have:

$$T(v) = \sum_{i=1}^n \alpha_i T(b_i)$$

Solution:

3.4.5. Prove that a linear map $T: V \rightarrow W$ is invertible $\iff T$ is injective and surjective.

Solution:

3.4.6. Let V and W be two vector spaces over a field $\mathbb{F}$ with dimensions n and $m > 3$ respectively. Construct three different linearly independent maps from V to W .

Solution:

3.4.7. Prove whether or not the triple $(\mathcal{L}(\mathbb{R}, \mathbb{R}), +, \cdot)$ forms a subring of $\mathcal{F}(\mathbb{R})$ under point-wise addition and multiplication.

Solution:

3.6.1. Let V and W be two vector spaces over $\mathbb{F}$ with $\dim(V) = n$ and $\dim(W) = m$. Then $\dim(\mathcal{L}(V, W)) = mn$

Solution:

3.6.2. Suppose A and B are two similar matrices of size n . Prove whether or not they represent, or can be viewed as matrix representations of the same linear transformation relative to suitable ordered bases.

Solution:

3.6.3. True/False. Justify

1. Let $\mathcal{B}$ and $\mathcal{B}'$ be two ordered bases for the vector space $\mathbb{R}^n$ over $\mathbb{R}$. Then $\det(\mathcal{M}_{\mathcal{B}}^{\mathcal{B}'}) = 1$
2. For all choices of ordered bases $\mathcal{B}$ and $\mathcal{B}'$ for $\mathbb{R}^n$, $\det(\mathcal{M}_{\mathcal{B}}^{\mathcal{B}'}) \neq 0$.
3. Let $T \in \mathcal{L}(V, W)$ be two vector spaces of dimension n over a field $\mathbb{F}$. If T is surjective, then it must be an isomorphism.
4. $\mathbb{P}_{10}[x] \cong \mathbb{R}^{11}$.
5. $T \in \mathcal{L}(V, W)$ is an isomorphism $\iff \dim(V) = \dim(W)$.
6. There are finite dimensional vector spaces V and W such that $\dim(\mathcal{L}(V, W)) = 32$.

Solution:

4.2.1 For a basic vector space V over $\mathbb{C}$, prove the following property of the inner product:

$$\forall u, v, w \in V, \langle v + u, w \rangle = \langle v, w \rangle + \langle u, w \rangle$$

Solution:

4.2.2 Verify the following function is an inner product on $\mathcal{C}_0([0, 1])$

$$\langle \cdot, \cdot \rangle : \mathcal{C}_0([0, 1]) \times \mathcal{C}_0([0, 1]) \rightarrow \mathbb{R}$$

$$\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x)dx$$

Solution:

4.2.3 Verify the following function is an inner product on $\mathbb{C}^n$

$$\vec{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_n), \quad \vec{\beta} = (\beta_1, \beta_2, \dots, \beta_n) \in \mathbb{C}^n$$

$$\langle \vec{\alpha}, \vec{\beta} \rangle = \sum_{i=1}^n \overline{\alpha_i} \beta_i$$

Solution:

4.2.4 Could we have defined the inner product on $\mathbb{C}^n$ by $\langle \vec{\alpha}, \vec{\beta} \rangle = \sum_{i=1}^n \alpha_i \overline{\beta_i}$. Would it be an inner product over $\mathbb{C}^n$?

Solution:

4.2.5. Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Then V is a normed vector space with norm $\|v\| = \sqrt{\langle v, v \rangle}$. Prove the positive-definiteness of the norm.

Solution:

4.2.6. Could we ever obtain equality in the Cauchy-Schwarz inequality? i.e.,

$$|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle} \sqrt{\langle v, v \rangle}$$

Solution:

4.2.7. Prove the generalised version of the Pythagorean Theorem. i.e., Let V be an inner product space. Let u, v be any two orthogonal vectors in V . Then:

$$\|u_v\|^2 = \|u\|^2 + \|v\|^2$$

Solution:

4.2.8. Define the discrete metric d on a vector space V as follows:

$$\forall u, v \in V, \quad d((u, v)) = \begin{cases} 0 & \text{if } u = v, \\ 1 & \text{otherwise.} \end{cases}$$

Verify that d is indeed a metric on V .

Solution:

4.2.9. In Example 4.2.9, we define a metric on $\mathcal{C}_0([0, 1], \mathbb{R})$ as follows:

$$d(f(x), g(x)) \longmapsto \int_0^1 |f(x) - g(x)| \psi(x) dx$$

(where $\psi : [0, 1] \rightarrow (0, \infty)$ is any fixed continuous function and $\psi \neq 1$)

Prove that function d is indeed a metric on $\mathcal{C}_0([0, 1], \mathbb{R})$. Make sure you understand it well. What happens if ψ is not continuous?

Solution:

4.2.10. Let $(V, \|\cdot\|)$ be a normed vector space. Also, let $\|\cdot\|$ follow the parallelogram law:

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$$

Define the inner product on V as:

$$(u, v) \longmapsto \langle u, v \rangle = \frac{1}{4} (\|u + v\|^2 - \|u - v\|^2 + i\|u - iv\|^2 - i\|u + iv\|^2)$$

Prove that $\langle \cdot, \cdot \rangle$ is indeed an inner product on V .

Solution:

4.3.1. The projection matrix is defined assuming that $A^\top A$ is invertible. We would be in big trouble if it's not! Luckily, $A^\top A$ turns out to be invertible. Can you show it?

Hint: $\text{Null}(A^\top A) = \text{Null}(A)$, and $A^\top A$ is a square matrix.

Solution:

4.4.1 [Equivalent Characterizations of Orthogonal Operators] Let V be a finite-dimensional real inner product space, and let $T : V \rightarrow V$ be a linear map. Then the following are equivalent.

1. T is an orthogonal operator.
2. $\langle T(v), T(w) \rangle = \langle v, w \rangle$ for all $v, w \in V$.
3. $\|T(v)\| = \|v\|$ for all $v \in V$.
4. T maps every orthonormal basis of V onto an orthonormal basis of V .
5. If $\hat{A}$ is the matrix representation of T with respect to any orthonormal basis of V , then

$$\hat{A}^T \hat{A} = \hat{I}$$

.

6. $T^{-1} = T^*$, where T^* denotes the adjoint of T .

Solution:

5.1.1. Prove properties of eigen vectors:

1. For any $k \in \mathbb{N}$, $A^k \vec{v} = \lambda^k \vec{v}$
2. Let $E_\lambda = \{0\} \cup \{\text{eigenvectors associated with } \lambda\}$, then E_λ is a subspace of the n -space! ($\mathbb{C}^n$ or $\mathbb{R}^n$)
 E_λ is called the eigenspace of λ

Solution:

5.3.1. Find and verify the spectral decomposition of $A = \begin{bmatrix} 3 & -2 \\ -2 & 0 \end{bmatrix}$

Solution:

5.3.2 Given $A \in M_{n \times n}(\mathbb{C})$, prove:

1. $(A^*)^* = A$
2. $(A + B)^* = A^* + B^*$
3. $\forall B \in M_{n \times n}(\mathbb{C}), (AB)^* = B^* A^*$

Solution:

5.3.3 Prove the eigenvalues of the Hermitian matrix are real.

Solution:

5.3.4. True or False: Suppose $A \in M_{n \times n}(\mathbb{C})$ is diagonalizable. Then A is Hermitian.

Solution:

5.3.5. Convince yourself that:

$$\widehat{U}_2 \text{ is unitary, and } \widehat{U}_2^* \widehat{C} \widehat{U}_2 = \left[\begin{array}{c|ccc} \lambda_1 & * & \cdots & * \\ \hline 0 & & & \\ \vdots & & \mathbf{B} & \\ 0 & & & \end{array} \right]$$

(Refer to proof of Schur's Lemma on page 162 for details on terms.)